

Assigned: {Assigned Date}

Due: {Due Date} before (Due Time)

Marks: 20 marks possible

Submission Procedure:

- **Create / debug / test your assignment in Visual Studio**
- Create a **ZIP** file of your CareerCloud solution and upload it to **BlackBoard**.
 - o Please ensure you upload the entire solution
 - o Name your ZIP file according to the guidelines in the '1 – Sundry' slide deck
 - E.g. NET.12-1-Joe-Smith-N#####.zip
- **DO NOT EMAIL YOUR ASSIGNMENT** – The Humber email server will not deliver the assignment and you will not be notified that it was not delivered.

Description:

In this assignment you will create class structures that represent database tables. These classes will be used to load data from the database into memory in later assignments. They will also be the foundation for the layered architecture we will build during the modules of our program.

Steps:

- Create a blank solution named 'CareerCloud'.
- Add a 'Class Library' project to the solution named 'CareerCloud.Pocos'. **The framework selected should be .NET 8.0 (Long Term Support).**
- Inside the 'CareerCloud.Pocos' project create a class for each table in the database. Follow the naming format: [table name] + 'Poco'
 - o All classes must be public
 - o For each table in the database there must be a corresponding class in this project
 - o Where the table name contains pluralization change it to be singular and remove the underscore characters. Example: Table – 'Applicant_Educations' would become 'ApplicantEducationPoco'.
- Inside each class in the Pocos project create the properties that correspond to the database columns
 - o All properties must be public
 - o Where a column name contains pluralization change it to be singular and remove the underscore characters. Example: Column – 'Company_Name' would become 'CompanyName'.
 - o Associate the corresponding data annotation (attribute) to the POCO classes and properties.

- Where a column is the primary key (or part of a primary key) place a [Key] attribute on that property.
- The POCO class must be decorated with the Table attribute. The table attribute will tell C# which class is mapped to which table in the database.
 - i.e. [Table("Applicant_Educations")] on the ApplicantEducationPoco class
- The POCO attributes must be decorated with the Column attribute if the property name is not the same as the column name. The column attribute will tell C# which property is mapped to which column in the database.
 - i.e. [Column("Certificate_Diploma")] on the ApplicantEducationPoco class's CertificateDiploma property.
- Create an interface in the 'CareerCloud.Pocos' project called IPoco
 - This interface must contain a single property, 'Id'.
 - The 'Id' property must be of type 'Guid' and have both a getter and a setter.
 - Apply the interface to all classes that correspond to tables that have an 'Id' column (and only those classes).

Testing Your Code:

You can test your code before submitting. There is a project on GitHub located here:

<https://github.com/johnhinz/CareerCloudCore.UnitTests.Assignment1.git>

Note: You are not to modify the unit test files to match your code. Your code should pass the unit tests without modifying the unit tests.

Follow these steps in order:

1. Clone the solution from GitHub using Visual Studio and the URI above.
2. Open the solution.
3. Build the test project
 - a. You should receive an error like the following (these errors are okay):

	Code	Description	Project
	CS0234	The type or namespace name 'Pocos' does not exist in the namespace 'CareerCloud' (are you missing an assembly reference?)	CareerCloud.Unit
	CS0234	The type or namespace name 'Pocos' does not exist in the namespace 'CareerCloud' (are you missing an assembly reference?)	CareerCloud.Unit
	CS0246	The type or namespace name 'ApplicantEducationPoco' could not be found (are you missing a using directive or an assembly reference?)	CareerCloud.Unit
	CS0246	The type or namespace name 'ApplicantEducationPoco' could not be found (are you missing a using directive or an assembly reference?)	CareerCloud.Unit

- b.
4. Copy the test project folder into your CareerCloud solution folder.
5. Add the project to your CareerCloud solution through Visual Studio

- a. In your assignments solution explorer window right click and select 'Add' -> 'Existing project'
 - b. Navigate to the 'CareerCloud.UnitTests.Assignment1.csproj' file you copied previously
6. Build the CareerCloud solution.
7. Open the Test Explorer window. (Menu: Test -> Windows -> Test Explorer)

You should now be able to run the unit tests on the assignment.

Reference:

Item	Link
Database Download	https://github.com/johnhinz/HumberDB.git
SQL types and corresponding C# types	https://msdn.microsoft.com/en-us/library/cc716729(v=vs.110).aspx
Attribute classes that are used to define metadata for ASP.NET MVC, ASP.NET data controls and EntityFramework	https://learn.microsoft.com/en-us/dotnet/api/system.componentmodel.dataannotations.schema?view=net-8.0 and https://learn.microsoft.com/en-us/dotnet/api/system.componentmodel.dataannotations?view=net-8.0
Working with Git	https://www.visualstudio.com/en-us/docs/git/tutorial/clone
Working with Unit Tests	https://www.visualstudio.com/en-us/docs/test/developer-testing/getting-started/getting-started-with-developer-testing